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Introduction

Lecture Scientific Practice

Overview

1. What does scientific research mean?

2. Literature research

Libraries and databases for the engineering sciences - Search techniques - Online search in free and fee-based databases - Content indexing of a library using the example of the University of Applied Sciences Nordhausen - Dealing with tesaurs

3. Technical standards

Objectives and procedures of technical standardization - National and international standards boards - Researching and reading technical standards

4. Patents and industrial property rights

Objectives and procedures in industrial property protection – patents, utility models, trademarks and designs - national and international patent organizations – German employee invention law - searching and reading patents

5. Writing of academic texts and lectures

Objectives and structure - Literature references - Lecture structure - Presentation techniques - Examples of bad practice

Forms of teaching

Work load: 2,5 ECTS



Attend the lecture with 2 SWS



Editing of a term paper. The term paper will be graded

In this lecture you will learn everything you need for your term paper or master thesis.



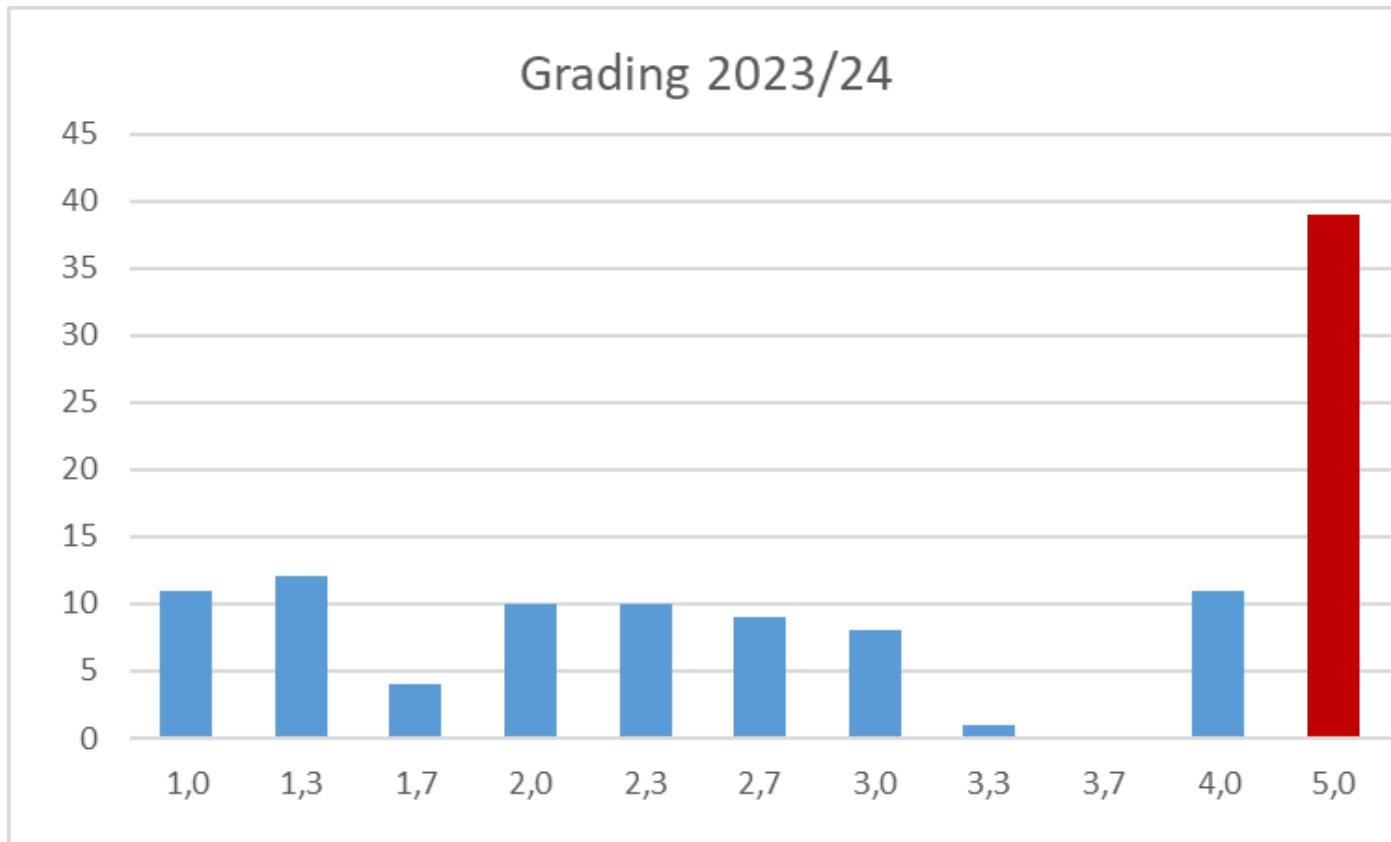
There is no attendance obligation!



There is no claim to a good grade if you ignore lecture contents in your seminar paper!

Forms of teaching

Results from the last year



What does scientific research mean?

Definition

1. On the basis of scientific knowledge and the state of scientific discussion in a subject area
2. engage with the scientific ideas of others,
3. to make your own thoughts,
4. do these in an understandable and transparent form and
5. in the spirit of the scientific community.

What does scientific research mean?

State of the scientific discussion



State of the art

- Standards



State of science and research

- Specialist books
- Journal articles
- Conference contributions
- Patent applications

What does scientific research mean?

Requirements for scientific work - Technical requirements

Objectivity: Objectivity is the name given to a process in which the result is independent of the person performing it.

Reproducibility: Does running the experiment again lead to the same result?

Reference to facts, i.e. observable processes and phenomena that can be described by word, measure or number. These must be distinguished from one's own opinions:

Example



Opinion: In my opinion, the portions in the Mensa are too large; conversations with my fellow students lead me to assume that the majority of students share this opinion.



Facts: According to a survey conducted by the Studentenwerk, 55% of the students think that the food portions are too large.

What does scientific research mean?

Requirements for scientific work - Formal requirements

- ➡ Respect of the applicable language rules
- ➡ Correct use of the respective technical language
- ➡ Structure and scope should facilitate discussion and access to the scientific content
- ➡ Correct citation

What does scientific research mean?

Correct citation - citation forms

A distinction is made between two forms of quotation:

Literal quotations consist in the literal adoption of a statement. This must be placed in inverted commas and concluded by a reference to the source.

"All our knowledge is patchwork." [Luther, 1534]

In the case of **analogous quotations**, however, the statement is reproduced with the same content but in your own words. In this case, too, it is necessary to cite the source in the immediate vicinity-

For example, MartinLuther [Luther, 1534] already pointed out that our knowledge still has considerable gaps.

or

There are also considerable gaps in our knowledge in many areas [cf. Luther, 1534].

What does scientific research mean?

Correct citation - citation methods

There are two different ways of citation:

One way of citing – sometimes called German citation method - is to use footnotes. A text mark is attached to the statement to be substantiated which refers to a footnote (usually on the same page) in which the complete source is mentioned. This method of citation is very common in the humanities.¹

In the Harvard citation method, only a reference is added to the statement, which refers to the complete source given in the bibliography. The reference can be a number [1], a name [Wesselak] or a name with a year [Wesselak 2019]. The Harvard citation method is mainly used in the technical sciences.

[1] Wesselak, Viktor: Lecture notes Scientific Practice, Nordhausen (2019)

¹ Wesselak, Viktor: Lecture Notes Scientific Practice, Nordhausen (2019)

What does scientific research mean?

Correct citation – buiding the bibliography

books:

surname(s), first name(s): title, place of publication, year of publication, [page(s)].

journal articles:

surname(s), first name(s): title of the article, title of the journal, [volume of the journal,] year of publication, issue number, page(s)

grey literature:

surname(s), first name(s): title, type of publication, source, year of publication, [page(s)].

What does scientific research mean?

Correct citation – buiding the bibliography

personal messages:

surname, first name: occasion, date

internet sources:

surname(s), first name(s): title, type of publication, source, URL, date of request

But beware:

If a statement is found on the internet, this does not mean that it is automatically an internet source!

What does scientific research mean?

Requirements for scientific work - moral standards

- ➡ truthfulness
... that starts with every single measurement!
- ➡ no plagiarism
- ➡ ethical and moral principles